AGENTIC SYSTEMS

Two loops correcting one value at different rates will fight.

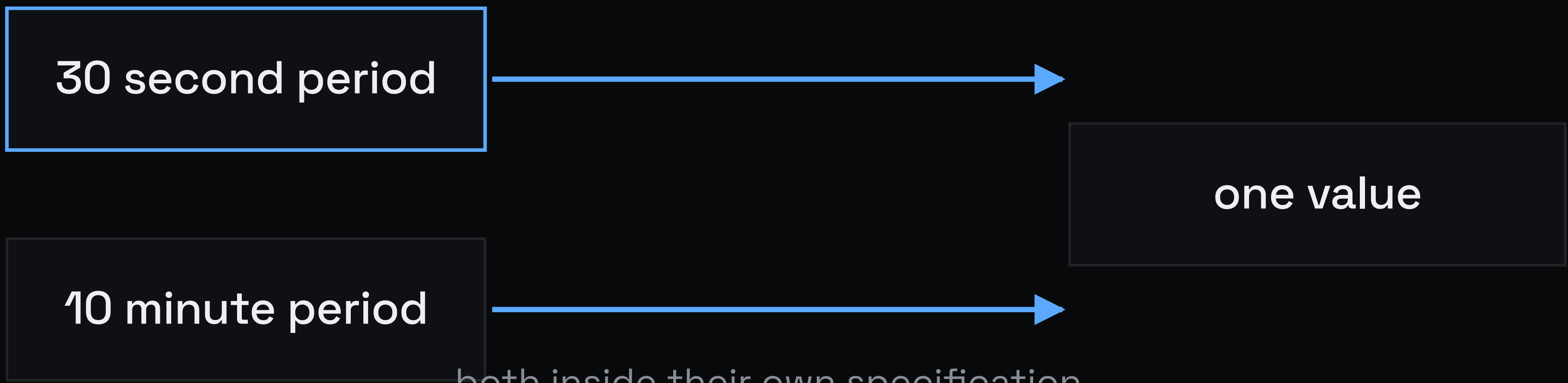
and both will report healthy while doing it

Contention.

Two processes want one resource, so they are ordered. That is a real problem with a settled answer.

WHAT ACTUALLY HAPPENS

Two controllers act on the same variable at different periods.



both inside their own specification

THE MECHANISM

The fast loop corrects. The slow loop overcorrects.

One samples every thirty seconds and applies a small correction. The other samples every ten minutes and applies a large one, computed from a state the fast loop has already partly corrected.

WHY IT IS INVISIBLE

Neither controller is malfunctioning.

Each operates inside its own specification, so each reports healthy, and the oscillation appears in neither set of metrics.

Nothing is waiting for anything.

There is no ordering to impose, because the conflict is not over access to a resource.

Separates the timescales.

Loops acting on the same variable are placed at rates far enough apart, commonly an order of magnitude, that the fast loop settles before the slow loop samples.

WHY THAT WORKS

It turns coordination into arithmetic.

The guarantee becomes that one controller always observes a world the other has finished settling.

Some loops require the same rate.

Separation is then unavailable, and the correct answer is one controller rather than two.

THE TEST

How far apart are their periods?

Take any two loops that touch the same value and compare. The tell is available without instrumentation.